

Solutions

Assignment 3

1) Given a system of linear equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

a necessary condition for convergence of Gauss-Seidel or Jacobi method is

$$\sum_{j \neq i} \left| \frac{a_{ij}}{a_{ii}} \right| < 1 \quad \text{for each } i=1,2,\dots,n$$

where strict inequality MUST hold (< 1) for AT LEAST one $i=1,2,\dots,n$

Systems satisfying this condition are called diagonally dominant.

2) Rearranging to make system diagonally dominant

$$x^{(n+1)} = \frac{1}{10} (26 + 2y^{(n)} + 5z^{(n)})$$

$$y^{(n+1)} = \frac{-1}{10} (39 - x^{(n)} - 3z^{(n)})$$

$$z^{(n+1)} = \frac{1}{10} (47 - 4x^{(n)} + 5y^{(n)})$$

Jacobi method

Given $(x^{(0)}, y^{(0)}, z^{(0)}) = (1, 1, 1)$

x	y	z
3.300000	-3.500000	4.800000
4.300000	-2.130000	1.630000
2.989000	-2.981000	1.915000
2.961300	-3.026600	2.013900
3.001630	-2.999700	2.002180
3.001150	-2.999183	1.999498
2.999912	-3.000036	1.999949
2.999967	-3.000024	2.000017
3.000004	-2.999998	2.000001
3.000001	-2.999999	1.999999
3.000000	-3.000000	2.000000
3.000000	-3.000000	2.000000

$$x = 3$$

$$y = -3$$

$$z = 2$$

3) Rearranging to make system diagonally dominant

$$x^{(n+1)} = \frac{1}{10} (9 - 2y^{(n)} - z^{(n)})$$

$$y^{(n+1)} = \frac{1}{10} (-22 - x^{(n)} + z^{(n)})$$

$$z^{(n+1)} = \frac{1}{10} (22 + 2x^{(n)} - 3y^{(n)})$$

Jacobi
method

Given $(x^{(0)}, y^{(0)}, z^{(0)}) = (1, -1, 1)$

x	y	z
1.000000	-2.200000	2.700000
1.070000	-2.030000	3.060000
1.000000	-2.001000	3.023000
0.997900	-1.997700	3.000300
0.999510	-1.999760	2.998890
1.000063	-2.000062	2.999830
1.000029	-2.000023	3.000031
1.000002	-2.000000	3.000013
0.999999	-1.999999	3.000000
1.000000	-2.000000	2.999999
1.000000	-2.000000	3.000000
1.000000	-2.000000	3.000000

$$x = 1$$

$$y = -2$$

$$z = 3 //$$

8 iterations to get accuracy to 5 significant figures //

4) Rearranging to make system diagonally dominant

$$x^{(n+1)} = \frac{1}{4} (-8 + 2y^{(n)} - z^{(n)})$$

$$y^{(n+1)} = \frac{1}{9} (11 - 3x^{(n+1)} + 2z^{(n)})$$

$$z^{(n+1)} = \frac{1}{13} (24 - 4x^{(n+1)} + 2y^{(n+1)})$$

Gauss Seidel
method

Let $(x^{(0)}, y^{(0)}, z^{(0)}) = (0, 0, 0)$

x	y	z
-2.000000	1.888889	2.752137
-1.743590	2.415005	2.754182
-1.481043	2.327944	2.660005
-1.501029	2.313677	2.663959
-1.509151	2.317264	2.667010
-1.508121	2.317598	2.666745
-1.507887	2.317461	2.666652
-1.507932	2.317456	2.666665

$$x = -1.507$$

$$y = 2.317$$

$$z = 2.666 //$$

5) The system is diagonally dominant

$$x^{(n+1)} = \frac{1}{6.7} (20.5 - 1.1y^{(n)} - 2.2z^{(n)})$$

$$y^{(n+1)} = \frac{1}{9.4} (22.9 - 3.1x^{(n+1)} + 1.5z^{(n)})$$

$$z^{(n+1)} = \frac{1}{8.4} (28.8 - 2.1x^{(n+1)} + 1.5y^{(n+1)})$$

Gauss Seidel
method

$$\text{Let } (x^{(0)}, y^{(0)}, z^{(0)}) = (0, 0, 0)$$

x	y	z
3.059701	1.427120	2.918489
1.867088	2.286145	3.370039
1.577784	2.453609	3.472270
1.516722	2.490060	3.494045
1.503587	2.497867	3.498722
1.500770	2.499542	3.499726
1.500165	2.499902	3.499941

$$x = 1.50$$

$$y = 2.49$$

$$z = 3.49 //$$

6) Rewriting the system as $AX = b$ we have

$$[A|B] = \left[\begin{array}{cccc|c} 1.660 & 0.684 & 0.820 & 0.380 & -4.925 \\ 0.784 & 1.690 & 1.396 & 0.492 & 6.105 \\ 0.754 & 1.602 & 1.608 & 0.456 & 7.325 \\ 0.442 & 0.570 & 0.338 & 1.398 & -4.175 \end{array} \right]$$

Steps of Gauss Jordan elimination

$$\left[\begin{array}{cccc|c} 1 & 0.412 & 0.493 & 0.228 & -2.966 \\ 0.784 & 1.69 & 1.396 & 0.492 & 6.105 \\ 0.754 & 1.602 & 1.608 & 0.456 & 7.325 \\ 0.442 & 0.57 & 0.338 & 1.398 & -4.175 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0.412 & 0.493 & 0.228 & -2.966 \\ 0 & 1.366 & 1.008 & 0.312 & 8.431 \\ 0 & 1.291 & 1.235 & 0.283 & 9.562 \\ 0 & 0.387 & 0.119 & 1.296 & -2.863 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0.412 & 0.493 & 0.228 & -2.966 \\ 0 & 1 & 0.737 & 0.228 & 6.167 \\ 0 & 1.291 & 1.235 & 0.283 & 9.562 \\ 0 & 0.387 & 0.119 & 1.296 & -2.863 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0.189 & 0.134 & -5.508 \\ 0 & 1 & 0.737 & 0.228 & 6.167 \\ 0 & 0 & 0.282 & -0.011 & 1.597 \\ 0 & 0 & -0.166 & 1.208 & -5.255 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0.189 & 0.134 & -5.508 \\ 0 & 1 & 0.737 & 0.228 & 6.167 \\ 0 & 0 & 1 & -0.041 & 5.652 \\ 0 & 0 & -0.166 & 1.208 & -5.255 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0.142 & -6.581 \\ 0 & 1 & 0 & 0.259 & 1.996 \\ 0 & 0 & 1 & -0.041 & 5.652 \\ 0 & 0 & 0 & 1.201 & -4.314 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0.142 & -6.581 \\ 0 & 1 & 0 & 0.259 & 1.996 \\ 0 & 0 & 1 & -0.041 & 5.652 \\ 0 & 0 & 0 & 1 & -3.591 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -6.069 \\ 0 & 1 & 0 & 0 & 2.929 \\ 0 & 0 & 1 & 0 & 5.502 \\ 0 & 0 & 0 & 1 & -3.591 \end{array} \right]$$

Ans

$$\begin{aligned} x &= -6.069 \\ y &= 2.929 \\ z &= 5.502 \\ u &= -3.591 \end{aligned}$$

7) Rewriting the system as $AX = b$ we have

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -1 & 3 & 3 \\ 4 & 2 & -2 & 10 \\ 6 & -3 & 1 & 9 \end{array} \right] \quad \text{where } X = \begin{bmatrix} \sin x \\ \cos y \\ \tan z \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 3 \\ 4 & 2 & -2 & 10 \\ 6 & -3 & 1 & 9 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \downarrow$$

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 3 \\ 0 & 4 & -8 & 4 \\ 0 & 0 & -8 & 0 \end{array} \right]$$

$$\Rightarrow \begin{array}{l} \sin x = 2 \\ \cos y = 1 \\ \tan z = 0 \end{array}$$

But $\sin x = 2$ has no solution

Thus the given system of linear equations has no real solutions. //

8) Given $P_2 = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$

Let $f(x) = ax^2 + bx + c \in P_2$

such that $f(1) = 2$ & $f(-1) = 6$

$$\Rightarrow \begin{aligned} a + b + c &= 2 \\ &\& a - b + c = 6 \end{aligned}$$

Rewriting the system as

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 6 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1 \quad \downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & 0 & 4 \end{array} \right]$$

$$R_2 \rightarrow \frac{R_2}{-2} \quad \downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & -2 \end{array} \right]$$

$$\Rightarrow b = -2 \quad \& \quad a + b + c = 2 \quad \text{i.e.} \quad c = 4 - a$$

$$\text{Hence, } f(x) = ax^2 - 2x + (4 - a) = 0 //$$

9) Rearranging to make system diagonally dominant

$$x^{(n+1)} = \frac{1}{3} (6 - y^{(n)} - z^{(n)})$$

$$y^{(n+1)} = \frac{1}{5} (10 - x^{(n)} - 3z^{(n)})$$

$$z^{(n+1)} = \frac{1}{7} (16 - 2x^{(n)} - 3y^{(n)})$$

Jacobi
method

$$x^{(n+1)} = \frac{1}{3} (6 - y^{(n)} - z^{(n)})$$

$$y^{(n+1)} = \frac{1}{5} (10 - x^{(n+1)} - 3z^{(n)})$$

$$z^{(n+1)} = \frac{1}{7} (16 - 2x^{(n+1)} - 3y^{(n+1)})$$

Gauss Seidel
method

Given $(x^{(0)}, y^{(0)}, z^{(0)}) = (1, 1, 1)$

Gauss Seidel

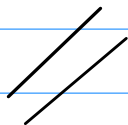
Jacobi

1.333333	1.133333	1.419048	1.333333	1.200000	1.571429
1.149206	0.918730	1.563628	1.076190	0.790476	1.390476
1.172547	0.827314	1.596138	1.273016	0.950476	1.639456
1.192183	0.803881	1.600570	1.136689	0.761723	1.514649
1.198516	0.799955	1.600443	1.241209	0.863873	1.634493
1.199867	0.799761	1.600141	1.167211	0.771062	1.560852
1.200033	0.799909	1.600030	1.222695	0.830047	1.621770
1.200020	0.799978	1.600004	1.182728	0.782399	1.580638
1.200006	0.799997	1.600000	1.212321	0.815071	1.612478
1.200001	0.800000	1.600000	1.190817	0.790049	1.590021
1.200000	0.800000	1.600000	1.206644	0.807824	1.606889
1.200000	0.800000	1.600000	1.195096	0.794538	1.594749
1.200000	0.800000	1.600000	1.203571	0.804132	1.603742
1.200000	0.800000	1.600000	1.197375	0.797041	1.597209
1.200000	0.800000	1.600000	1.201917	0.802200	1.602018
1.200000	0.800000	1.600000	1.198594	0.798406	1.598510
1.200000	0.800000	1.600000	1.201028	0.801175	1.601085
1.200000	0.800000	1.600000	1.199247	0.799143	1.599202
1.200000	0.800000	1.600000	1.200551	0.800629	1.600582

Ans $x = 1.2$

$y = 0.8$

$z = 1.6$



- 10) No, the given equation cannot be made to be diagonally dominant by any rearrangement. Hence, neither the Jacobi nor Gauss Seidel method will converge for the given system.

- 11) No, the given system of linear equations is not diagonally dominant. In fact, no rearrangement of the equations will make the system diagonally dominant.

12) Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$

The solution of $AX = B$ is given by $X = A^{-1}B$ (if A is invertible)

Now to find A^{-1} we use Gauss Jordan method to reduce the matrix $[A|I]$ to $[I|C]$ where I is the 3×3 identity matrix. Then $C = A^{-1}$.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 4 & 3 & -1 & 0 & 1 & 0 \\ 3 & 5 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -5 & -4 & 1 & 0 \\ 0 & 2 & 0 & -3 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\substack{R_1 \rightarrow R_1 + R_2 \\ R_3 \rightarrow R_3 + 2R_2}} \left[\begin{array}{ccc|ccc} 1 & 0 & -4 & -3 & 1 & 0 \\ 0 & -1 & -5 & -4 & 1 & 0 \\ 0 & 0 & -10 & -11 & 2 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow -R_2 + \frac{R_3}{2} \\ R_1 \rightarrow R_1 - 4R_3 \\ R_3 \rightarrow -R_3/10}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{5} & \frac{1}{5} & -\frac{2}{5} \\ 0 & 1 & 0 & -\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{11}{10} & -\frac{1}{5} & -\frac{1}{10} \end{array} \right]$$

$$\text{Hence } A^{-1} = \begin{bmatrix} \frac{7}{5} & \frac{1}{5} & -\frac{2}{5} \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ \frac{11}{10} & -\frac{1}{5} & -\frac{1}{10} \end{bmatrix} \Rightarrow X = A^{-1}B = \begin{bmatrix} 1 \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$(x, y, z) = (1, \frac{1}{2}, -\frac{1}{2}) //$$

13)

	0	0				
			0.12			
	0.2	0.12		0.24		
			0.36		-0.98	
	0.4	0.48		-0.74		4.0
			-0.38		3.02	-10.0
x_0	0.6	0.10	2.28	-6.0		
			1.90		-2.98	
	0.8	2.00		-0.70		
			1.20			
	1.0	3.20				

Difference
Table

As $t = 0.7$ lies in center of table, using Stirling's formula (taking $x_0 = 0.6$)

$$y = y_0 + p \left(\frac{\Delta y_{-1} + \Delta y_0}{2} \right) + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2-1)}{3!} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{p^3(p^2-1)}{4!} \Delta^4 y_{-2}$$

where $p = \frac{x - 0.6}{0.2}$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{h} \left(\frac{\Delta y_{-1} + \Delta y_0}{2} + p \Delta^2 y_{-1} + \frac{3p^2-1}{12} (\Delta^3 y_{-2} + \Delta^3 y_{-1}) + \frac{2p^3-p}{12} \Delta^4 y_{-2} \right)$$

$$x = 0.7 \Rightarrow p = 0.5$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{x=0.7} = 10.12083 //$$

14)

1.2	0.9320				
		0.0316			
1.3	0.9636		-0.0097		
		0.0219		-0.0002	
1.4	0.9855		-0.0099		0.0002
		0.0120		0.0000	
1.5	0.9975		-0.0099		
		0.0021			
1.6	0.9996				

Difference
Table

Using Newton's forward formula terminated to $\Delta^3 y_0$ we have

$$\Delta y_0 + \frac{2p-1}{2} \Delta^2 y_0 + \frac{3p^2-6p+2}{6} \Delta^3 y_0 = 0$$

$$\text{where } p = \frac{x-1.2}{0.1}$$

$$0.316 + \frac{2p-1}{2} (-0.0097) + \frac{3p^2-6p+2}{6} (-0.0002) = 0$$

$$\Rightarrow -0.0001p^2 - 0.0095p + 0.311083 = 0$$

$$\Rightarrow p = -120.76 \text{ or } 25.7604$$

$$\Rightarrow x = -10.876 \text{ or } x = 3.77604$$

Ignoring negative value we have maximum occurs at $x = 3.77604$

Using Newton's Backwards interpolation or Lagrange's formula,

$$\text{maximum value} = 1.02184$$

$$\text{at } x = 3.77604 //$$

15)

x_0	0.0	3.013					
	0.1	3.162	0.149				
	0.2	3.207	0.045	-0.104	0.216		
	0.3	3.364	0.157	0.112	-0.238	-0.454	
	0.4	3.395	0.031	-0.126	0.319	0.773	-1.171
	0.5	3.381	-0.014	0.081	-0.398		
	0.6	3.324	-0.043	-0.079			
			-0.057	0.002			

Difference
Table

Stirling's Formula

$$\begin{aligned}
 y = y_0 &+ p \left(\frac{\Delta y_{-1} + \Delta y_0}{2} \right) + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2-1)}{3!} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \\
 &+ \frac{p^2(p^2-1)}{4!} \Delta^4 y_{-2} + \frac{p(p^2-1)(p^2-4)}{5!} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \\
 &+ \frac{p^2(p^2-1)(p^2-4)}{6!} \Delta^6 y_{-3}
 \end{aligned}$$

where $p = \frac{x - x_0}{h}$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} = \frac{1}{h} &\left(\frac{\Delta y_{-1} + \Delta y_0}{2} + p \Delta^2 y_{-1} + \frac{(3p^2-1)}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \right. \\
 &+ \frac{(2p^3-p)}{12} \Delta^4 y_{-2} + \frac{(5p^4-15p^2+4)}{120} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \\
 &\left. + \frac{(3p^5-10p^3+4p)}{360} \Delta^6 y_{-3} \right)
 \end{aligned}$$

$$\& \frac{d^2 y}{dx^2} = \frac{1}{h^2} \left(\Delta^2 y_{-1} + p \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{6p^2 - 1}{12} \Delta^4 y_{-2} \right. \\ \left. + \frac{2p^3 - 3p}{12} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \frac{15p^4 - 20p^2 + 4}{360} \Delta^6 y_{-3} \right)$$

At $x = 0.3$ we have $p = 0$

Hence

$$\left. \frac{dy}{dx} \right|_{x=0.3} = \frac{1}{h} \left(\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \right. \\ \left. + \frac{1}{30} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \right)$$

$$\& \left. \frac{d^2 y}{dx^2} \right|_{x=0.3} = \frac{1}{h^2} \left(\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \frac{1}{90} \Delta^6 y_{-3} \right)$$

Substituting values we have

(a) Velocity at $0.3s = 1.1333 //$

(b) Acceleration at $0.3s = -16.5594 //$

16)

	0	135				
	300	149	14			
			8	24		
	600	157	18		-50	
			26	-26	70	
x_0	900	183	-8	20	-86	
			18	-6	-16	
	1200	201	-14	4		
			4	-2		
	1500	205	-16			
			-12			
	1800	193				

Difference
Table

Using Stirling formula as in previous question

$$\left. \frac{dy}{dx} \right|_{x=900} = 0.08522 //$$

17)

50	73499		
		-775	
51	72724	-196	
		-971	13
52	71753	-183	
		-1154	
53	70599		

Difference
Table
(Take $y = lx$)

Newton's Forward formula

$$y = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$$

where $p = \frac{x-50}{1}$ i.e. $\frac{dp}{dx} = 1$

$$\Rightarrow \frac{dy}{dx} = \Delta y_0 + \frac{2p-1}{2} \Delta^2 y_0 + \frac{3p^2-6p+2}{6} \Delta^3 y_0$$

At $x=50$, $p=0$

$$\Rightarrow \frac{dy}{dx} \Big|_{x=50} = \Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3}$$

$$= -672.6667$$

$$\Rightarrow \mu_x = \frac{-1}{l_x} \frac{dl_x}{dx} \quad \text{at } x=50$$

$$\text{is } \frac{-1}{73499} \times (-672.6667)$$

$$= 0.009152 //$$

18)

0.4	1.583649				
		0.213793			
0.5	1.797443		0.033002		
		0.246795		0.003471	
x_0 0.6	2.044238		0.036473		0.000365
		0.283268		0.003836	
0.7	2.327505		0.040309		
		0.323576			
0.8	2.651082				

Difference
Table

Using Stirling formula as in previous questions

$$\frac{dy}{dx} \Big|_{x=0.6} = 2.6442258$$

$$f'(x) = 2e^x - 1 \Rightarrow f'(0.6) = 2.6442376$$

$$\text{Error} = 0.00018 //$$

19)

1920	250			
		35		
1925	285		8	
		43		65
1930	328		73	
		116		
1940	444			

Difference Table

From Newton's Backward Formula we have

$$y = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n$$

$$\text{where } p = \frac{x - 1940}{5} \Rightarrow \frac{dp}{dx} = \frac{1}{5}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{5} \left(\nabla y_n + \frac{2p+1}{2} \nabla^2 y_n + \frac{3p^2+6p+2}{6} \nabla^3 y_n \right)$$

$$\text{At } x = 1935, p = -1$$

$$\begin{aligned} \Rightarrow \left. \frac{dy}{dx} \right|_{x=1935} &= \frac{1}{5} \left(116 - \frac{1}{2} \times 73 - \frac{1}{6} \times 65 \right) \\ &= 13.7333 \end{aligned}$$

20) The values in this question are too big for scientific calculators to handle. Hence, as an illustration of the method integration from 0.5 to 0.6 is shown here.

$$x_0 = 0.5, x_{12} = 0.6 \Rightarrow h = \frac{x_{12} - x_0}{12} = \frac{1}{120}$$

Hence we have

x	y	
0.500000	0.841468	Using Trapezoidal Rule 0.841466075220361
0.508333	0.841468	
0.516667	0.841468	
0.525000	0.841467	
0.533333	0.841467	Using Simpson's 1/3-rd Rule 0.841466079340766
0.541667	0.841467	
0.550000	0.841466	
0.558333	0.841466	
0.566667	0.841465	Using Simpson's 3/8-th Rule 0.841466079335527
0.575000	0.841465	
0.583333	0.841464	
0.591667	0.841464	
0.600000	0.841463	Using Weddle's Rule 0.841466079344958

Trapezoidal Rule

$$\int_{x_0}^{x_n} y dx = \frac{h}{2} \left(y_0 + y_n + 2 \sum_{i=1}^{n-1} y_i \right)$$

Simpson's $1/3$ Rule

$$\int_{x_0}^{x_n} y dx = \frac{h}{3} \left(y_0 + y_n + 4 \sum_{i \text{ odd}} y_i + 2 \sum_{\text{Rest}} y_i \right)$$

Simpson's $3/8$ Rule

$$\int_{x_0}^{x_n} y dx = \frac{3h}{8} \left(y_0 + y_n + 2 \sum_{i=3k} y_i + 3 \sum_{\text{Rest}} y_i \right)$$

Weddle's Rule

$$\int_{x_0}^{x_n} y dx = \frac{3h}{10} \left(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6 + y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12} \right)$$

21) $x_0 = 0$, $x_{10} = \frac{5}{3} \Rightarrow h = \frac{x_{10} - x_0}{10} = \frac{1}{6}$

x	y
0.000000	0.000000
0.166667	0.142857
0.333333	0.250000
0.500000	0.333333
0.666667	0.400000
0.833333	0.454545
1.000000	0.500000
1.166667	0.538462
1.333333	0.571429
1.500000	0.600000
1.666667	0.625000

Using Trapezoidal Rule

0.683854340104340

Ans 0.684

22) $x_0 = 0$, $x_n = 2$, $h = 0.25$

$\Rightarrow n = \frac{x_n - x_0}{h} = 8$

x	y
0.250000	-0.009407
0.500000	-0.037629
0.750000	-0.052709
1.000000	0.000000
1.250000	0.189277
1.500000	0.594308
1.750000	1.302532
2.000000	2.408240

As $n=8$ we can use Trapezoidal or Simpson's $\frac{1}{3}$ Rule only

Using Trapezoidal Rule

distance = 0.798799130164111

Using Simpson's $\frac{1}{3}$ -rd Rule

distance = 0.625312628418562

23) As $n=10$ we can use Trapezoidal or Simpson's $1/3$ Rule only

x	$y = A/\sqrt{x}$
1.500000	1.026336
1.650000	1.082114
1.800000	1.132941
1.950000	1.181590
2.100000	1.248329
2.250000	1.308000
2.400000	1.370391
2.550000	1.437185
2.700000	1.498325
2.850000	1.569725
3.000000	1.632169

By trapezoidal rule

$$T = 1.973677627970073$$

By Simpson's $1/3$ -rd Rule

$$T = 1.973646320178541 //$$

24) For Weddle's Rule $n = 6k$, so taking $n = 6$ or 12 or 18
 $x_0 = 0$, $x_{12} = \frac{\pi}{2} \Rightarrow h = \frac{x_n - x_0}{n}$

$$n = 6, h = \frac{\pi}{12}$$

x	y
0.000000	1.000000
0.261799	0.994559
0.523599	0.979541
0.785398	0.958645
1.047198	0.937283
1.308997	0.921332
1.570796	0.915423

Using Weddle's Rule

$$1.505103337732758 //$$

$$n = 12, \quad h = \pi/24$$

x	y
0.000000	1.000000
0.130900	0.998619
0.261799	0.994559
0.392699	0.988067
0.523599	0.979541
0.654498	0.969518
0.785398	0.958645
0.916298	0.947647
1.047198	0.937283
1.178097	0.928291
1.308997	0.921332
1.439897	0.916930
1.570796	0.915423

Using Weddle's Rule

1.505103382105331

//

$$n = 18, \quad h = \pi/36$$

x	y
0.000000	1.000000
0.087266	0.999385
0.174533	0.997555
0.261799	0.994559
0.349066	0.990479
0.436332	0.985427
0.523599	0.979541
0.610865	0.972987
0.698132	0.965953
0.785398	0.958645
0.872665	0.951280
0.959931	0.944085
1.047198	0.937283
1.134464	0.931093
1.221730	0.925716
1.308997	0.921332
1.396263	0.918088
1.483530	0.916095
1.570796	0.915423

Using Weddle's Rule

1.505103382105388

//

The End